Credit Risk Scorecard — Home Credit Indonesia

Predicting payment difficulty to approve the right applicants

The challenge
8.07% of approved loans ran into payment difficulty, yet every rejected good applicant costs growth and financial inclusion.

The objective
Rank applicants by the probability of payment difficulty and turn it into an approve / review / decline policy.

The expected value
A less risky approved book, explainable decisions and a policy whose cut-off follows from real margins and losses.

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Dataset Overview

Eight linked tables, reduced to one row per applicant

307,511
labelled loans for model development

48,744
unlabelled loans to score

58.1 M
rows of bureau, previous-loan and repayment history

8.07%
payment-difficulty rate: about 1 loan in 12

Key data-quality findings and how they were treated

Finding	Treatment
18.0% of loans carry a coded “365243” employment value (pensioners, unemployed)	Set to missing, plus a not-employed flag
41 of 122 application columns are over 50% missing, and missingness signals risk	Kept, with missing indicators
Instalment data holds one row per payment, not per instalment	Payments summed per instalment first
Monthly bureau status covers 35% of training vs 99.98% of scoring applicants	Excluded as a drift risk

Exploratory Analysis

Risk concentrates in a few clear signals

External credit scores
19.92%
difficulty rate in the lowest score decile vs **3.26%** in the highest

Card utilisation
19.92%
above 90% utilisation vs **5.29%** with no balance

Prior refusals
15.90%
when half or more of past applications were refused vs **7.05%** never refused

Age
12.44%
at ages 20–25 vs **4.91%** at 60–70, falling steadily with age

Counter-intuitive: loan size relative to income barely ranks risk
Credit-to-income has almost no ranking power (AUC 0.502), and its risk is lowest at both extremes. A simple debt-to-income cut-off would target the wrong applicants; financing above the goods price is the better affordability signal.

Data Preparation and Feature Engineering

Repayment history adds real predictive value

1 Aggregate, then join
103 engineered features: affordability ratios plus bureau, previous-loan, POS, card and instalment history, one row per applicant.

2 Leakage-safe pipelines
Imputation, winsorising, log transforms, binning and encoding are fitted inside each training fold only; a test confirms it.

3 Quality fixes found by testing
Unreported bureau annuities had been summed to zero; fixed. Features reported very differently in scoring data are excluded.

+0.0216 ROC-AUC
from adding the history tables (0.7579 → 0.7795), better in all 5 folds; every table contributes something the others don't

25.73%
difficulty rate after 90+ days late **in the last 12 months**, vs 9.57% for 90+ days at any time: recent behaviour matters most

Modelling Approach

Two model families, compared under rules fixed in advance

Development · 80% · 5 stratified folds
All selection, tuning and comparison

Holdout · 20%
Locked; opened once

Scoring set
Predictions only

Logistic Regression (challenger)
Interpretable scorecard baseline. Imputation with missing flags, winsorising, log transforms, decile bins for non-linear ratios. Tuning the L2 strength changed almost nothing (+0.0001).

Gradient boosting (selected)
Captures non-linear effects and interactions; handles missing values and categories natively. Balanced class weights, then isotonic recalibration. Tuning added about +0.004.

Decisions by pre-set rules: external-score summaries removed · 55 zero-importance features kept (removal hurt the linear model) · undersampling rejected, SMOTE not used · gradient boosting chosen only if ≥ +0.005 ROC-AUC in ≥ 4 of 5 folds.

Model Performance

Gradient boosting ranks risk best, in validation and on the holdout

Finding	Log. Reg. CV	Gr. boost CV	Log. Reg. holdout	Gr. boost holdout
ROC-AUC (ranking)	0.7778	0.7846	0.7827	0.7879
Gini	0.5556	0.5691	0.5654	0.5759
PR-AUC (minority class)	0.2645	0.2745	0.2748	0.2818
KS statistic	0.4194	0.4328	0.4332	0.4357
Brier score (lower is better)	0.0667	0.0662*	0.0662	0.0656
Calibration error (ECE)	0.0023	0.0005*	0.0034	0.0028

CV = mean of 5 development folds
** after cross-validated isotonic recalibration*
Holdout used once

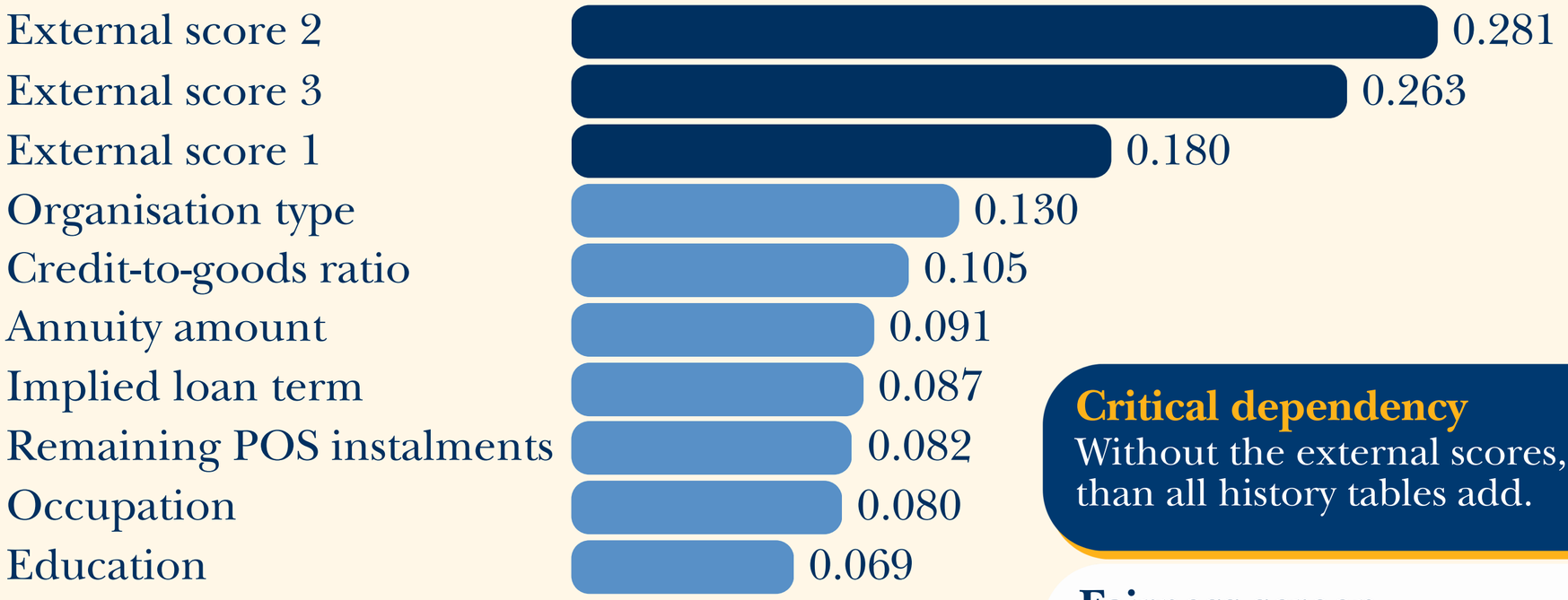
+0.0068
ROC-AUC over Logistic Regression, better in 5 of 5 folds: passes the pre-set selection rule

At a reference threshold
Flagging 13.6% of holdout applicants catches 45.6% of difficulty cases; about 1 in 4 flagged applicants actually has difficulties.

Model Explainability

External scores dominate; loan structure and history follow

Average impact on predicted risk (mean |SHAP|, 400 holdout applicants)



Explanations validated
The standard SHAP tool failed an exactness check on this model; an exact method replaced it. Directions match Logistic Regression for 17 of 18 features.

Critical dependency
Without the external scores, ROC-AUC falls by 0.026, more than all history tables add.

Fairness screen
Similar ranking across gender and age; small calibration gaps (women +0.38 pp, men −0.74 pp) for compliance review.

Decision Threshold and Business Impact

A three-band policy makes the approved book safer

Band (predicted probability)	Applicants	Difficulty rate	Share of all difficulty cases
Approve below 0.144	84.06%	4.90%	51.00%
Review 0.144 to 0.25	10.21%	18.84%	23.84%
Decline 0.25 and above	5.73%	35.45%	25.16%

8.07% → 6.41%
difficulty rate among approved loans when declining at 0.25; holdout confirms 6.32%

+2.85%
profit vs approving everyone, in an **illustrative** scenario (10% margin, 30% loss)

The cut-off is a business choice: decline when risk ≥ margin ÷ (margin + loss). The data confirm it (best threshold 0.2514 vs 0.25 in theory), so Finance and Risk only need to supply real margins and losses.

Recommendations and Roadmap

Pilot first, then deploy with monitoring

1 Set the cut-off from real economics
Finance and Risk supply margin and loss; threshold = margin ÷ (margin + loss).

2 Verify, don’t auto-reject, the review band
Check income, employment and bureau exposure, or offer a lower limit; never decline on one feature.

3 Use recent repayment history
Keep instalment and bureau data reliable at decision time; fix bureau reporting gaps.

4 Govern the external scores
Document provenance and availability; keep a fallback model ready.

Before launch
Compliance review of age as an input, the calibration gaps and reason-code wording.

Pilot
Champion–challenger on a share of live applications, with a small controlled sample approved below the threshold.

Rollout and monitoring
Monthly score stability (alert at PSI 0.1) and band mix; retrain if calibration drifts over 1 pp or ROC-AUC drops 0.02.

Conclucion

A reliable, explainable scorecard, ready for a controlled pilot

Performance
Holdout ROC-AUC 0.788, well-calibrated probabilities, stable score on new applications (PSI 0.0066).

Value
Difficulty rate in the approved book falls from 8.07% to about 6.4%, with a review band for borderline cases.

Limitations
Early-difficulty target, approved applicants only, no out-of-time test, reliance on external scores.

References

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